2026.08.02 - Supervised Learning Notation (slide version)

Notation matches `paper/nature-biotech/sections/methods.tex` (Problem setup, Eq. 1–2, and Table 1 “Notation”). Supersedes the tilde notation $(\tilde{G}, \tilde{E}, \tilde{P})$ in [[scratch.2025.07.07.183123-torchcell-basic-supervised-formulation]]. Rendered one slide per page at 16:9, into the usual PDF output directory:

```
cd notes
F=scratch.2026.08.02.235304-supervised-learning-notation
pandoc -s "$F.md" -o "assets/pdf-output/$F.pdf" \
  --pdf-engine=tectonic -M title="" \
  -V documentclass=extarticle -V fontsize=20pt -V pagestyle=empty \
  -V geometry:'paperwidth=13.333in,paperheight=7.5in,margin=1in'
```

The learning problem

$$\hat{f}_\theta : \ \mathcal{G} \times \mathcal{E} \times \mathcal{P} \ \longrightarrow \ \mathcal{Y}$$

$$\hat{\Theta} = \arg\min_{\Theta} \ \mathbb{E}_{(G,\varepsilon,p,y)\sim D}[\mathcal{L}(\hat{f}_\theta(G,\varepsilon,p),\ y)]$$

Where

- $G = (N, E)$: cell graph on $N = 6607$ gene nodes, relation-k adjacency $A^{(k)} \in \{0, 1\}^{N \times N}$
- ε: environment (media, temperature)
- $p = \{(e_t, \tau_t, m_t)\}_{t=1}^{M}$: perturbation set – entity, type, magnitude
- $\mathcal{G},\ \mathcal{E},\ \mathcal{P},\ \mathcal{Y}$: cell-graph, environment, perturbation, and phenotype spaces
- $y \in \mathcal{Y}$: observed phenotype; $\widehat{y}$ predicted
- D: data law over (G, ε, p, y)
- $\mathcal{L}$: loss function
- $\Theta = (\theta, \psi, \phi)$: learnable parameters

The model factors: encode, perturb, decode

$$\hat{f}_\theta(G,\varepsilon,p) = \underbrace{\mathcal{R}_\phi}_{\text{DEC}}\left(\ \underbrace{\mathcal{T}_\psi}_{\text{PERT}}\left(\underbrace{F_\theta(G,\varepsilon)}_{\text{ENC}},\ p\right)\right)$$

- F_θ (ENC): cell encoder, $H = F_\theta(G,\varepsilon)$
- $\mathcal{T}_\psi$ (PERT): perturbation operator, $H_{\text{pert}} = \mathcal{T}_\psi(H,p)$
- $\mathcal{R}_\phi$ (DEC): decoder, $\hat{y} = \mathcal{R}_\phi(H_{\text{pert}})$

The environment enters the **encoder**: H represents this cell *in this condition*, so the perturbation operator acts on a condition-aware encoding and can change it accordingly. Abundant entity data trains F_θ; the scarce phenotype labels need only fit $\mathcal{T}_\psi$ and $\mathcal{R}_\phi$.

Layman version (same content, non-ML audience)

Alternate slides for a talk where the audience is biology-first. Same symbols, no ML vocabulary: nothing here says “loss”, “distribution”, “parameters”, or “objective”.

What we are actually asking for

We want one function that takes a **cell**, a **growth condition**, and a **perturbation** (a change to the genome), and tells us **what the cell does**. Training means turning the model's internal dials until its predictions sit as close as possible to what we measured in the lab.

$$\underbrace{\hat{\Theta}}_{\substack{\text{the dial settings}\\\text{we keep}}} = \arg\min_{\Theta} \underbrace{\mathbb{E}_{(G,\varepsilon,p,y)\sim D}}_{\text{averaged over every experiment}} \Big[\underbrace{\mathcal{L}}_{\substack{\text{how far off}\\\text{we were}}} \big(\underbrace{\hat{f}_\theta(G,\varepsilon,p)}_{\text{the prediction}}, \underbrace{y}_{\text{the measurement}} \big) \Big]$$

What goes in

- G – **the cell we start from.** Wild-type yeast: its 6607 genes, and the known links between them (regulation, physical interaction, metabolism).
- ε – **the growth condition.** Which medium, what temperature.
- p – **the perturbation**, the change we make to the genome. A list, each entry saying *which* gene (e_t), *what kind* of change (τ_t: deleted, overexpressed, ...), and *how much* (m_t). Deleting two genes is a list of length two.

What comes out, and how we grade it

- y – **what we measured** in the lab for that strain (e.g. how fast it grew). $\hat{y}$ is what the model predicted instead.
- D – **the pile of real experiments** we learn from: every (cell, condition, perturbation, result) record we have.
- $\mathcal{L}$ – **the scorecard.** One number saying how far the prediction was from the measurement. Training pushes it down.
- Θ – **the dials.** All the internal numbers the model is free to adjust; there is one group per stage below.
- $\mathcal{G}, \mathcal{E}, \mathcal{P}, \mathcal{Y}$ – **"every possible one"** of the above: every cell, every condition, every change, every phenotype. The curly letters just mean we are talking about the whole range, not one example.

How the model is put together

$$\hat{f}_\theta(G,\varepsilon,p) = \underbrace{\mathcal{R}_\phi}_{\substack{\text{3. report}\\\text{the phenotype}}} \Big(\underbrace{\mathcal{T}_\psi}_{\substack{\text{2. apply the}\\\text{perturbation}}} \Big(\underbrace{F_\theta(G,\varepsilon)}_{\substack{\text{1. read the cell}\\\text{in its condition}}}, p\Big)\Big)$$

Reading the cell is the expensive step, and we do it **once** per condition. The perturbation is then applied to that reading rather than to the cell, so millions of strains reuse a single reading. Because the reading already knows the growth condition, the same deletion is free to matter in one medium and not in another. That is the whole trick: the plentiful data (genomes, networks) pays for step 1, and the scarce data (measured strains) only has to pay for steps 2 and 3.

What changed from the old slide

Old	New	Why
$\tilde{\mathcal{G}}$, $\tilde{G}$	$\mathcal{G}$, $G = (N, E)$	tildes dropped; $\tilde{A}^{(k)}$ now means row-normalized adjacency
$\tilde{\mathcal{E}}$, $\tilde{E}$	$\mathcal{E}$, ε	E is the edge set of G; ε frees it
$\tilde{\mathcal{P}}$, $\tilde{P}$	$\mathcal{P}$, p	$\mathbf{P}$ is the perturbation token matrix; p is the perturbation set
$\tilde{P}$ = "perturbation operator"	$\mathcal{T}_\psi$ = perturbation operator	the operator is the learned map, not the data
$\hat{\theta} = \arg\min_\theta$	$\hat{\Theta} = \arg\min_\Theta$, $\Theta = (\theta, \psi, \phi)$	one symbol for all three factors' parameters

Also changed here, relative to the current `methods.tex` draft (Eq. 2): the environment ε moves from the decoder into the encoder, $\mathcal{R}_\phi(\mathcal{T}_\psi(F_\theta(G), p), \varepsilon) \Rightarrow \mathcal{R}_\phi(\mathcal{T}_\psi(F_\theta(G, \varepsilon), p))$, so that PERT operates on a condition-aware encoding. **The manuscript has not been updated to match** – see the handoff note below.

Paper locations still carrying ε in the decoder

- `sections/methods.tex:24` – Eq. 2 (`eq:factor`), status `todo`
- `sections/methods.tex:76-80` – Table 1 rows for F_θ, $\mathcal{R}_\phi$, status `todo`
- `sections/methods.tex:267,283` – operator/amortization paragraphs, status `todo`
- `sections/backmatter.tex:86,144,148` – Supplementary Note on functional equivalence, status **`tent`** (author-approved; needs explicit go-ahead before editing)

Amortization argument under the change: encode-once becomes encode-once **per (reference, condition) pair**. Conditions are few and strains are $\sim 10^{4\text{–}6}$, so "few references, many perturbations" survives, but the wording in Methods and in the Supplementary Note should say so explicitly rather than implying one encoding per genome.